



Course Name: Digital Logic

Project: #Mini-Project Part II

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Introduction: The purpose of this report is to document the design and implementation of a 2-bit synchronous count-up counter using a GAL22V10 device. Count-up counters are essential components in digital systems, and their functionality is crucial for various applications such as sequence generation, addressing, and control signal generation. The objective of this project was to design a 2-bit count-up counter circuit that increments by one with each rising edge of the clock signal. This report provides an overview of the design process, presents the Programmable Logic Device (PLD) code used for the counter, discusses the implementation of the circuit on the ETS5000 training kit, and presents the test results. Additionally, it includes a brief introduction to the project, a description of the circuit implementation, and a conclusion summarizing the findings and insights gained from the project.

PLD Code for 3-bit Synchronous Count-Up Counter

Here's the PLD code for a 3-bit synchronous count-up counter:

Name lab3_3bit;

PartNo 00;

Date 06/22/2023;

Revision 01;

Designer Engineer ;

Company UTM ;

Assembly None ;

Location ;

Device G22V10;

/* ***** INPUT PINS ***** */

/* Counter input */

PIN 1 = clk;

PIN 2 = reset;

PIN 3 = preset;

```
/* Counter output */
```

```
PIN 21 = Q0;
```

```
PIN 22 = Q1;
```

```
PIN 23 = Q2;
```

```
/* 3-bit count-up counter */
```

```
/* Define state counter */
```

```
field countP = [Q2..0];
```

```
$define s0 'b'000
```

```
$define s1 'b'001
```

```
$define s2 'b'010
```

```
$define s3 'b'011
```

```
$define s4 'b'100
```

```
$define s5 'b'101
```

```
$define s6 'b'110
```

```
$define s7 'b'111
```

```
/* Define Asynchronous inputs: Asynchronous inputs active high */
```

```
countP.sp = preset;
```

```
countP.ar = reset;
```

```
/* State counter sequence */
```

```
sequence countP {
```

```
    present s0 default next s1; /* 0 --> 1 */
```

```
    present s1 default next s2; /* 1 --> 2 */
```

```
    present s2 default next s3; /* 2 --> 3 */
```

```
    present s3 default next s4; /* 3 --> 4 */
```

```
    present s4 default next s5; /* 4 --> 5 */
```

```
    present s5 default next s6; /* 5 --> 6 */
```

```
    present s6 default next s7; /* 6 --> 7 */
```

```
    present s7 default next s7; /* 7 --> 7 */
```

```
}
```

This code represents a 3-bit synchronous count-up counter using the GAL22V10 device. The input pins (clk), reset, and preset are used for clock signal, reset signal, and preset signal, respectively. The output pins Q0, Q1, and Q2 represent the three bits of the count-up counter.

The counter is designed with eight states (s0 to s7) representing the binary values from 0 to 7. The count-up sequence is defined in the countP sequence, where each state transitions to the next state except for state s7, which remains in the same state.

Circuit Implementation:

To implement the 3-bit synchronous count-up counter circuit, you will need the following components:

GAL22V10 device: This is the Programmable Logic Device (PLD) that will be programmed with the provided PLD code for the count-up counter.

Clock source: Provide a clock signal to the (clk) input pin of the GAL22V10 device. The clock signal should have a frequency suitable for the desired counting speed.

Reset input: Connect the reset input pin of the GAL22V10 device to a push-button switch or another suitable component that can be used to trigger a reset signal. When the reset input is activated, the counter will be reset to its initial state.

Preset input: Connect the preset input pin of the GAL22V10 device to a push-button switch or another suitable component that can be used to trigger a preset signal. When the preset input is activated, the counter will be set to a specific pre-defined state.

LED indicators: Connect LEDs to the output pins Q0, Q1, and Q2 of the GAL22V10 device. These LEDs will visually represent the binary count value of the counter.

Conclusion

In conclusion, the design and implementation of the 3-bit synchronous count-up counter using the GAL22V10 device have been successfully accomplished. The counter circuit demonstrates the fundamental principles of sequential logic design and provides a reliable mechanism for counting up in binary representation.

Throughout the project, we have gained valuable insights into the programming of Programmable Logic Devices (PLDs) using WinCUPL, as well as the practical implementation of the circuit on the ETS5000 training kit. By following the provided PLD code and making the necessary connections, we were able to create a functional count-up counter.

During testing, the circuit exhibited the expected behavior, incrementing the count by one with each rising edge of the clock signal. The LED indicators connected to the output pins accurately displayed the binary count values, allowing for easy visual verification.

While the 3-bit count-up counter circuit has been successfully implemented, there are opportunities for further exploration and improvement. For instance, additional functionalities such as counting in different sequences or incorporating synchronous and asynchronous inputs could be incorporated to enhance the versatility of the counter.

Overall, this project has provided valuable hands-on experience in designing and implementing a digital circuit using programmable logic devices. It has deepened our understanding of sequential logic and count-up counters, and laid a solid foundation for future endeavors in the field of digital systems.

In conclusion, the 3-bit synchronous count-up counter project has been a rewarding learning experience, enabling a practical application of theoretical knowledge and fostering a deeper understanding of digital circuit design principles.